

Summations and geometric progressions



Outline

Summations

Geometric progressions



Summations

Introduction: Summations

- ▶ In this section we introduce the **summation symbol** $\sum$, a compact notation to write sums of many terms
- ▶ We will see its **six main properties** (product by a constant, constant term, union of summations, decomposition, index translation and index reflection)
- ▶ Finally, we will compute some **important summations**: the sum of the first n natural numbers, of the first n odd numbers and of the first n even numbers

The summation symbol

Definition 1: summation

Let $a_1, a_2, \dots, a_n$ be n real numbers. Their sum

$$a_1 + a_2 + \cdots + a_n$$

can be written in compact form with the summation symbol:

$$\sum_{k=1}^n a_k$$

which is read: “sum for k from 1 to n of a_k ”. The symbol k is called the summation index.

- ▶ The summation symbol is therefore just a shorthand, which is nevertheless very useful when the terms a_k are defined explicitly as a function of the index k .

Example

$$\sum_{k=1}^{10} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

$$\sum_{k=3}^n k^2 = 3^2 + 4^2 + 5^2 + \cdots + n^2$$

Dummy index

- ▶ The summation index is a **dummy index**. This means that if we replace k with i , j or any other index (in all its occurrences), the value of the summation does not change.

Example

We have:

$$\sum_{k=1}^n k^2 = \sum_{i=1}^n i^2$$

since both symbols denote the sum of the squares of the first n natural numbers (without zero).

On the other hand, we have:

$$\sum_{k=1}^n k^2 \neq \sum_{k=1}^m k^2$$

since the two symbols denote the sum of, respectively, the first n or the first m squares (if $n \neq m$ the result will be different).

Main properties of summations

We now see six fundamental properties of summations, which will help us to manipulate and simplify expressions involving the symbol $\sum$.

Product by a constant

Observation 1: product by a constant

Given a summation $\sum_{k=1}^n a_k$ and a real number $c \in \mathbb{R}$, we have:

$$\sum_{k=1}^n (c \cdot a_k) = c \sum_{k=1}^n a_k$$

Proof

By the distributive property we have:

$$\underbrace{c a_1 + c a_2 + \cdots + c a_n}_{=\sum_{k=1}^n (c \cdot a_k)} = \underbrace{c (a_1 + a_2 + \cdots + a_n)}_{=c \sum_{k=1}^n a_k}$$



Summation with a constant term

Observation 2: summation with a constant term

For every natural number $n \geq 1$ and every real number $c \in \mathbb{R}$, we have:

$$\sum_{k=1}^n c = c \cdot n$$

Proof

We have:

$$\underbrace{c + c + \cdots + c}_{= \sum_{k=1}^n c \quad \text{that is, } c \text{ added } n \text{ times}} = c n$$



Observation 3: union of summations

Given two summations $\sum_{k=1}^n a_k$ and $\sum_{k=1}^n b_k$, we have:

$$\sum_{k=1}^n a_k + \sum_{k=1}^n b_k = \sum_{k=1}^n (a_k + b_k)$$

Proof

We have:

$$\begin{aligned} & \underbrace{a_1 + a_2 + \cdots + a_n + b_1 + b_2 + \cdots + b_n}_{=\sum_{k=1}^n a_k + \sum_{k=1}^n b_k} \\ &= \underbrace{a_1 + b_1 + a_2 + b_2 + \cdots + a_n + b_n}_{=\sum_{k=1}^n (a_k + b_k)} \end{aligned}$$



- ▶ The three properties that follow (decomposition, index translation and index reflection) are simply different ways of writing and/or ordering the terms of the summations
- ▶ To prove them it is enough to write out the terms of the two summations and observe that they are the same

Observation 4: decomposition

Given two natural numbers $n \geq 1$ and $m \geq 1$, we have:

$$\sum_{k=1}^{n+m} a_k = \sum_{k=1}^n a_k + \sum_{k=n+1}^{n+m} a_k$$

Proof

The terms of the summation $\sum_{k=1}^{n+m} a_k$ can be grouped into the first n and the last m ones:

$$\begin{aligned} & \underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^n a_k} + \underbrace{a_{n+1} + a_{n+2} + \cdots + a_{n+m}}_{=\sum_{k=n+1}^{n+m} a_k} \\ &= \underbrace{a_1 + a_2 + \cdots + a_n + a_{n+1} + \cdots + a_{n+m}}_{=\sum_{k=1}^{n+m} a_k} \end{aligned}$$



Observation 5: index translation

Given a summation $\sum_{k=1}^n a_k$ and a natural number $m \geq 1$, we have:

$$\sum_{k=1}^n a_k = \sum_{k=1+m}^{n+m} a_{k-m} = \sum_{k=1-m}^{n-m} a_{k+m}$$

Proof (1/2)

In the summation $\sum_{k=1+m}^{n+m} a_{k-m}$ the index k ranges from $1+m$ to $n+m$, and therefore the index $k-m$ ranges from 1 to n :

$$\begin{aligned} & \underbrace{a_{(1+m)-m} + a_{(2+m)-m} + \cdots + a_{(n+m)-m}}_{=\sum_{k=1+m}^{n+m} a_{k-m}} \\ &= \underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^n a_k} \end{aligned}$$

Proof (2/2)

In the same way, in the summation $\sum_{k=1-m}^{n-m} a_{k+m}$ the index k ranges from $1 - m$ to $n - m$, and therefore the index $k + m$ ranges from 1 to n :

$$\begin{aligned} & \underbrace{a_{(1-m)+m} + a_{(2-m)+m} + \cdots + a_{(n-m)+m}}_{=\sum_{k=1-m}^{n-m} a_{k+m}} \\ &= \underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^n a_k} \end{aligned}$$



Observation 6: index reflection

Given a summation $\sum_{k=1}^n a_k$, we have:

$$\sum_{k=1}^n a_k = \sum_{k=1}^n a_{n-k+1} = \sum_{k=0}^{n-1} a_{n-k}$$

Proof (1/2)

In the summation $\sum_{k=1}^n a_{n-k+1}$ the index k ranges from 1 to n , and therefore the index $n - k + 1$ ranges from n to 1, that is, the terms are the same ones listed in reverse order:

$$\underbrace{a_n + a_{n-1} + \cdots + a_1}_{=\sum_{k=1}^n a_{n-k+1}} = \underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^n a_k}$$

Proof (2/2)

In the same way, in the summation $\sum_{k=0}^{n-1} a_{n-k}$ the index k ranges from 0 to $n-1$, and therefore the index $n-k$ ranges from n to 1:

$$\underbrace{a_n + a_{n-1} + \cdots + a_1}_{=\sum_{k=0}^{n-1} a_{n-k}} = \underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^n a_k}$$



Some important summations

Using the properties just seen, we now compute in closed form some important summations.

Sum of the first n natural numbers

Observation 7: sum of the first n natural numbers (without zero)

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Proof

Using the index reflection:

$$\begin{aligned}\sum_{k=1}^n k &= \frac{1}{2} \left(\sum_{k=1}^n k + \sum_{k=1}^n k \right) = \frac{1}{2} \left(\sum_{k=1}^n k + \sum_{k=1}^n (n - k + 1) \right) \\ &= \frac{1}{2} \sum_{k=1}^n (k + n - k + 1) = \frac{1}{2} \sum_{k=1}^n (n + 1) = \frac{n(n+1)}{2}\end{aligned}$$



Sum of the first n odd numbers

Observation 8: sum of the first n odd numbers

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n (2k - 1) = n^2$$

or equivalently

$$\sum_{k=0}^{n-1} (2k + 1) = n^2$$

Proof (1/2)

Using the properties of summations we have:

$$\begin{aligned}\sum_{k=1}^n (2k - 1) &= 2 \sum_{k=1}^n k - \sum_{k=1}^n 1 \\ &= 2 \left(\frac{n(n+1)}{2} \right) - n = n^2 + n - n = n^2\end{aligned}$$

Proof (2/2)

For the second form:

$$\begin{aligned}\sum_{k=0}^{n-1} (2k + 1) &= 2 \sum_{k=0}^{n-1} k + \sum_{k=0}^{n-1} 1 \\&= 2 \sum_{k=1}^n (k - 1) + n = 2 \left(\sum_{k=1}^n k - \sum_{k=1}^n 1 \right) + n \\&= 2 \left(\frac{n(n+1)}{2} - n \right) + n = n^2 + n - 2n + n = n^2\end{aligned}$$



Sum of the first n even numbers

Observation 9: sum of the first n even numbers (without zero)

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n 2k = n(n+1)$$

Proof

$$\sum_{k=1}^n 2k = 2 \sum_{k=1}^n k = 2 \left(\frac{n(n+1)}{2} \right) = n(n+1)$$



- ▶ The three important summations we have just seen can also be proved by induction, in the chapter “Principle of induction”
- ▶ Here we have obtained them using only the properties of summations

Summary

Summations:

- ▶ $\sum_{k=1}^n a_k$ is a compact notation for $a_1 + a_2 + \cdots + a_n$; the index k is a **dummy index**
- ▶ The six properties: product by a constant, constant term, union of summations, decomposition, index translation and index reflection

Important summations ($n \geq 1$):

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n (2k-1) = n^2, \quad \sum_{k=1}^n 2k = n(n+1)$$

Geometric progressions



Introduction: Geometric progressions

- ▶ In this section we introduce **geometric progressions**: sequences in which each term is obtained from the previous one by multiplying it by a constant, the **common ratio**
- ▶ We will derive the **closed formula of the k -th term** and the formula for the **sum of the first n terms** of a geometric progression
- ▶ This formula appears often in the course (for example in the study of geometric series)

Geometric progression

Definition 2: geometric progression

*A sequence of real numbers is a **geometric progression** if the ratio between each term (starting from the second one) and the previous one is constant. This constant is called the common ratio of the progression.*

Terms of a geometric progression

Given the first term $a \in \mathbb{R}$ and the common ratio $q \in \mathbb{R}$, the associated geometric progression is:

$$a, \quad a q, \quad a q^2, \quad a q^3, \quad a q^4, \quad \dots$$

Each term (starting from the second one) is obtained from the previous one by multiplying it by q . The k -th term ($k \in \mathbb{N}$, $k \geq 1$) can be written as $a q^{k-1}$ and we have:

$$a q^{1-1} = a q^0 = a \quad \text{first term, in position } k = 1$$

$$a q^{2-1} = a q^1 = a q \quad \text{second term, in position } k = 2$$

$$a q^{3-1} = a q^2 \quad \text{third term, in position } k = 3$$

$$a q^{4-1} = a q^3 \quad \text{fourth term, in position } k = 4$$

$\dots$

$\dots$

Observation 10: k -th term of a geometric progression

Given the first term $a \in \mathbb{R}$ and the common ratio $q \in \mathbb{R}$, let t_k denote the term in position k . The recursive relation that defines the progression is:

$$t_1 = a \quad \text{and} \quad t_k = t_{k-1} q \quad \text{for every } k \geq 2$$

and the closed formula of the k -th term is:

$$t_k = a q^{k-1} \quad \text{for every } k \geq 1$$

Proof

We apply repeatedly the recursive relation:

$$t_1 = a = a q^0$$

$$t_2 = t_1 q = a q = a q^1$$

$$t_3 = t_2 q = a q q = a q^2$$

$$t_4 = t_3 q = a q^2 q = a q^3$$

$$\vdots$$

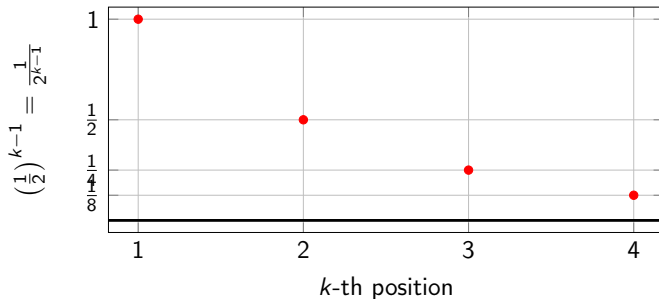
$$t_k = t_{k-1} q = a q^{k-2} q = a q^{k-1}$$



Example (1/2)

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

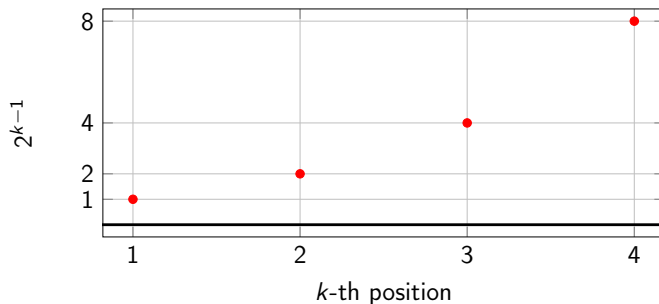
$$1, \quad \frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}$$



Example (2/2)

- With $a = 1$ and $q = 2$, the first 4 terms are:

1, 2, 4, 8



Sums of the terms of geometric progressions

We now see how to compute in closed form the sum of the first n terms of a geometric progression.

Sum of the first n terms

Observation 11: sum of the first n terms of the geometric progression ($a = 1$)

Given $q \in \mathbb{R}_+$, for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^n q^{k-1} = \begin{cases} \frac{q^n - 1}{q - 1} & \text{if } q \neq 1 \\ n & \text{otherwise} \end{cases}$$

Proof (1/2)

If $q \neq 1$, we prove the equation in the following equivalent form:

$$(q - 1) \sum_{k=1}^n q^{k-1} = q^n - 1$$

Applying the properties of summations, in particular the product by a constant and the index translation, we get:

$$\begin{aligned}(q - 1) \sum_{k=1}^n q^{k-1} &= q \sum_{k=1}^n q^{k-1} - \sum_{k=1}^n q^{k-1} = \\&= \sum_{k=1}^n q^k - \sum_{k=1}^n q^{k-1} = \sum_{k=1}^n q^k - \sum_{k=0}^{n-1} q^k = \\&= \sum_{k=1}^{n-1} q^k + q^n - \left(1 + \sum_{k=1}^{n-1} q^k \right) = q^n - 1\end{aligned}$$

Proof (2/2)

Dividing by $q - 1 \neq 0$ we obtain:

$$\sum_{k=1}^n q^{k-1} = \frac{q^n - 1}{q - 1}$$

If $q = 1$, we have instead:

$$\sum_{k=1}^n q^{k-1} = \sum_{k=1}^n 1 = n$$



Extension to the case $a \in \mathbb{R}$

- Given $q \in \mathbb{R}_+, n \in \mathbb{N}, n \geq 1$ and $a \in \mathbb{R}$, the formula extends as follows:

$$\sum_{k=1}^n a q^{k-1} = \begin{cases} a \left(\frac{q^n - 1}{q - 1} \right) & \text{if } q \neq 1 \\ a n & \text{otherwise} \end{cases}$$

since

$$\sum_{k=1}^n a q^{k-1} = a \sum_{k=1}^n q^{k-1}$$

Equivalent form

► Clearly:

$$\frac{q^n - 1}{q - 1} = \frac{1 - q^n}{1 - q}$$

and hence we also have:

$$\sum_{k=1}^n a q^{k-1} = \begin{cases} a \left(\frac{1-q^n}{1-q} \right) & \text{if } q \neq 1 \\ a n & \text{otherwise} \end{cases}$$

- ▶ The formula for the sum of the first n terms can also be proved by induction, in the chapter “Principle of induction”
- ▶ We now see three numerical examples, which check the formula with a direct computation

Example (1/3)

- With $a = 2$ and $q = \frac{1}{2}$, the first 4 terms are:

$$2, 1, \frac{1}{2}, \frac{1}{4} \quad \text{and} \quad 2 + 1 + \frac{1}{2} + \frac{1}{4} = \frac{8 + 4 + 2 + 1}{4} = \frac{15}{4}$$

the sum of the first $n = 4$ terms is given by the formula:

$$\begin{aligned} \sum_{k=1}^4 2 \left(\frac{1}{2} \right)^{k-1} &= \sum_{k=1}^4 \frac{2}{2^{k-1}} = 2 \frac{1 - \frac{1}{2^4}}{1 - \frac{1}{2}} \\ &= 2 \frac{1 - \frac{1}{16}}{\frac{1}{2}} = \frac{15}{16} \cdot 4 = \frac{15}{4} \end{aligned}$$

Example (2/3)

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

$$1, \quad \frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8} \quad \text{and} \quad 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{8 + 4 + 2 + 1}{8} = \frac{15}{8}$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^4 \left(\frac{1}{2}\right)^{k-1} = \sum_{k=1}^4 \frac{1}{2^{k-1}} = \frac{1 - \frac{1}{2^4}}{1 - \frac{1}{2}} = \frac{1 - \frac{1}{16}}{\frac{1}{2}} = \frac{15}{16} \cdot 2 = \frac{15}{8}$$

Example (3/3)

- With $a = 1$ and $q = 2$, the first 4 terms are:

$$1, 2, 4, 8 \quad \text{and} \quad 1 + 2 + 4 + 8 = 15$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^4 2^{k-1} = \frac{2^4 - 1}{2 - 1} = 16 - 1 = 15$$

Summary

Geometric progressions:

- ▶ First term a , common ratio q : terms $a, aq, aq^2, \dots$
- ▶ Recursive relation $t_1 = a, t_k = t_{k-1} q$; closed formula $t_k = a q^{k-1}$

Sum of the first n terms:

- ▶ If $q \neq 1$: $\sum_{k=1}^n a q^{k-1} = a \frac{q^n - 1}{q - 1} = a \frac{1 - q^n}{1 - q}$
- ▶ If $q = 1$: $\sum_{k=1}^n a q^{k-1} = a n$

Final summary

- ▶ The symbol $\sum$ allows us to write sums of many terms in compact form; its index is a dummy index
- ▶ The six properties of summations (product by a constant, constant term, union, decomposition, index translation and index reflection) allow us to compute sums in closed form
- ▶ Important summations:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n (2k-1) = n^2, \quad \sum_{k=1}^n 2k = n(n+1)$$

- ▶ In a geometric progression with common ratio $q \neq 1$:

$$\sum_{k=1}^n a q^{k-1} = a \frac{q^n - 1}{q - 1}$$